CSAI3121 · GROUP PROJECT

Next-Day Weather Prediction for Macau

t+1

Forecasting tomorrow's temperature and weather condition with machine learning

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Macao Polytechnic University · November 2026

Two tasks, one day ahead

01 Regression

Tomorrow's maximum and minimum temperature (°C), one model each

Metric: MAE

02 Classification

Tomorrow's condition: Clear, Partly Cloudy, Cloudy or Rain

Metric: Macro F1

Data source

15

days in the suggested Visual Crossing file: a forecast, not history

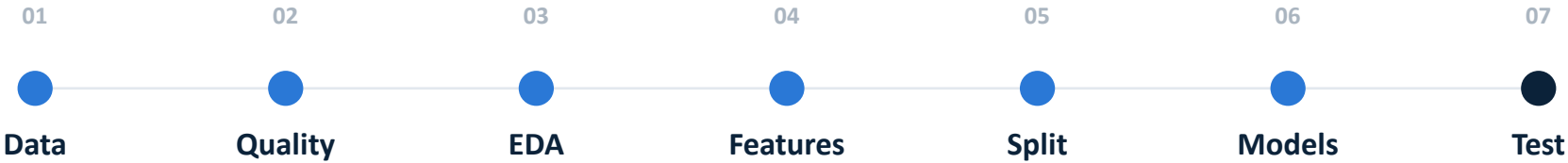
→ **4,383**

days of ERA5 history from Open-Meteo, 2014–2025

- Same 18 columns as the Visual Crossing file
- Quality audit with physical rules: 0 issues
- Windiest days match real typhoons

METHOD

How the code is organised



`weather_prediction_macau.ipynb`

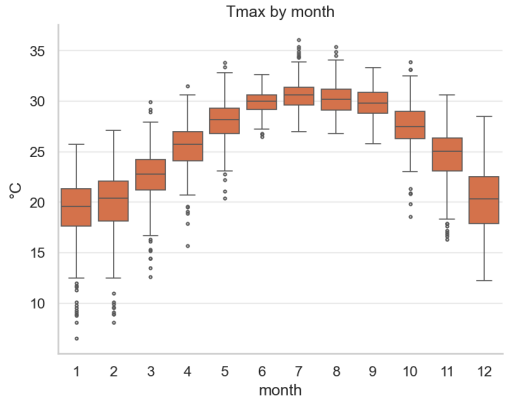
- The analysis step by step, with an interpretation after each result
- About 470 lines; runs end to end with Restart & Run All
- Fixed seed: every run reproduces the same tables

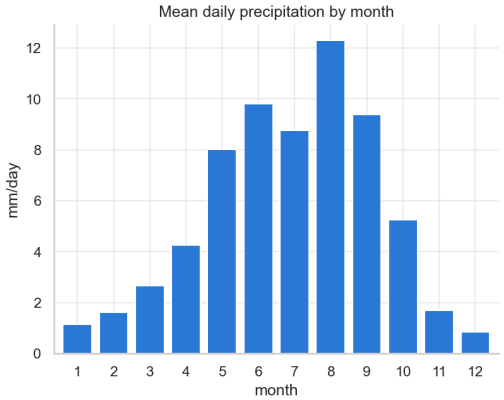
`weather_utils.py`

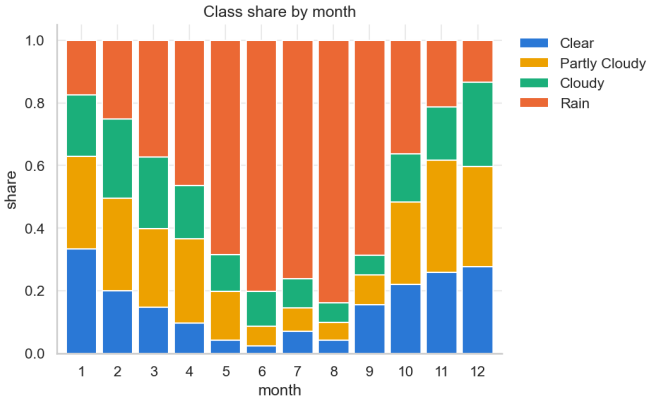
- Data: download, format conversion, coverage check
- Quality and 76 features
- 9 regressors, 8 classifiers, grids, ensemble
- Metrics and block bootstrap

EXPLORATORY ANALYSIS

Strong memory and strong seasons







0.93

lag-1 autocorrelation of Tmax: persistence is a hard baseline

>69%

of days are rainy from May to September

48%

of all days are Rain, so we use macro F1 instead of accuracy

FEATURE ENGINEERING

76 features, all built from day t or earlier

14
Today's values
current state of the atmosphere

20
Lags
t−1 and t−2: warming or cooling?

5
Differences
1-day and 3-day changes

24
Rolling statistics
3, 7 and 14-day windows

3
Physical
diurnal range, dew-point depression

4
Seasonality
sine and cosine of day and month

2
Wind direction
sine and cosine, so 359° ≈ 1°

4
Today's class
one-hot persistence

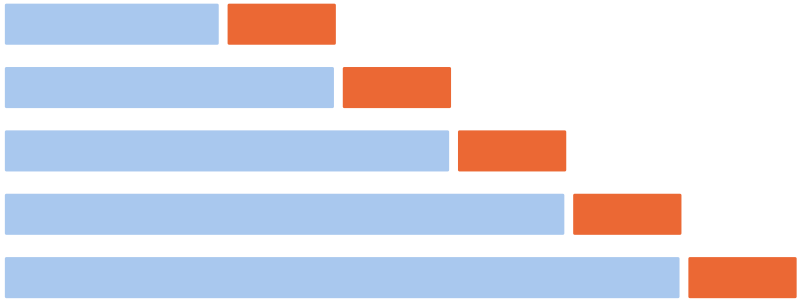
Leakage checks pass: target(t) equals Tmax(t+1) in 100% of rows, and no feature copies the target (max |r| = 0.939).

Evaluation respects time



2014–2021 · 2,892 days | 2022–23 · 730 days | 2024–25 · 731 days

Grid search over five time-ordered folds

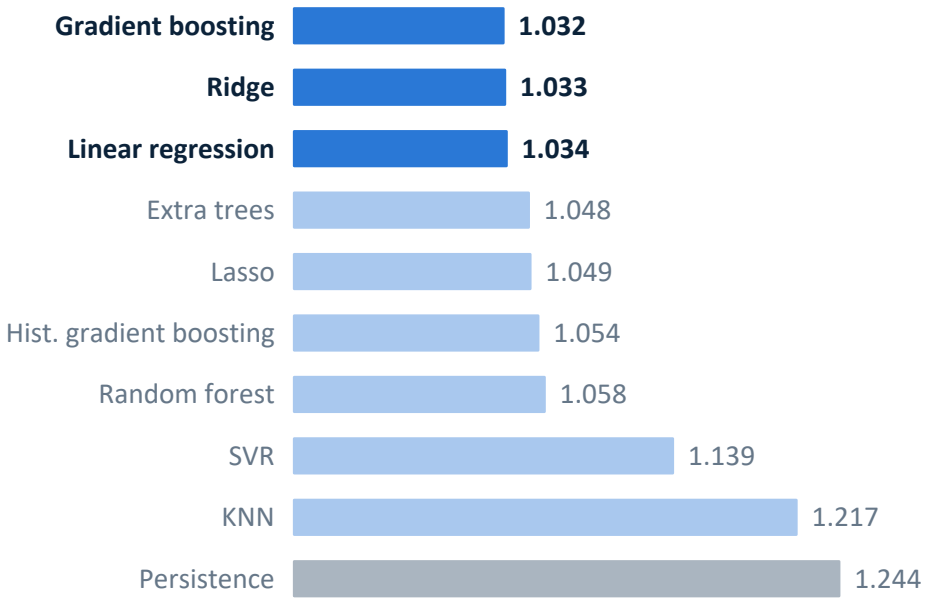


fit on the past score the next period

- Split by the date being predicted: no future label reaches training
- A shuffled k-fold would train on the future and inflate scores
- Final model chosen on validation, then tested once
- 7-day block bootstrap tests significance

Nine regressors: the top three are within 0.002 °C

Validation MAE for tomorrow's Tmax (°C) · axis starts at 0.9



Linear ≈ boosting

Ridge trains in milliseconds, gradient boosting in about 6 s.

Tuning: no gain

Grid search moved MAE from 1.032 to 1.035 °C.

Top-3 average wins

Boosting + Ridge + linear regression: 1.011 °C.

19% and 11% less error than persistence

1.22 °C

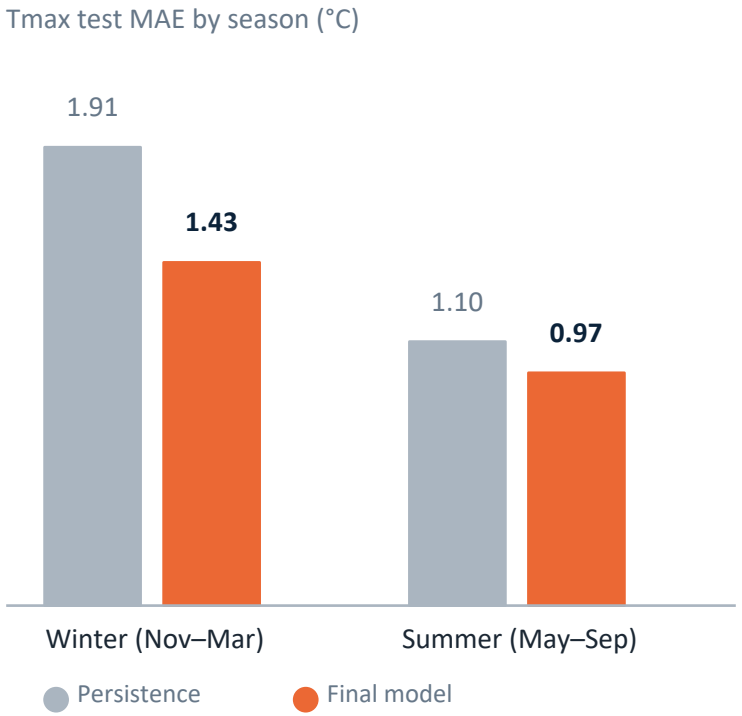
Tmax MAE, vs 1.51 °C for persistence

0.99 °C

Tmin MAE, vs 1.11 °C for persistence

Both gains are statistically significant

95% CI of the MAE reduction: [0.21, 0.36] °C for Tmax and [0.07, 0.17] °C for Tmin. The model still smooths extremes, so the largest misses fall on the sharpest changes.



Close to persistence, but useful rain odds

0.501

macro F1 on test, vs 0.489 for persistence (not significant)

0.88

ROC AUC of the rain probability

- Rain is predicted well (F1 0.81)
- The three dry classes are split by cloud thresholds and get confused
- Recall of Clear rises from 0.49 to 0.63
- Brier score 37% better than climatology



Row-normalised test confusion matrices: persistence (left), final ensemble (right)

DISCUSSION

What made the difference

01 Model diversity

Gradient boosting and extra trees were not in our first three models, yet won. Tmax test MAE: 1.27 → 1.22 °C.

02 Ensembles over tuning

Averaging different families helped; grid search barely moved the scores.

03 Features carry the signal

Today's values, rolling means and seasonality dominate the importance.

Limitations and responsible use

- One grid point: approaching typhoons and fronts are invisible
- Classes depend on our thresholds (1 mm, 30%, 70%)
- Only two test years
- Always report uncertainty (about ±1 °C, rain as a probability)
- Never a substitute for SMG warnings

CONCLUSION

Good features and diverse models beat complexity

A simple baseline is hard to beat in daily weather. We beat it clearly for temperature and report honestly where we did not.

–19%
Tmax error

–11%
Tmin error

0.88
rain AUC

Questions?